

Which Arab? How LLMs Rate Bank Customers from the 22 Arab League Countries in Money-Laundering Checks

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1 Research questions

Language models have been shown to carry stereotypes about Arabs and Muslims (Section 2). The studies that show this treat them as one group. The 22 Arab League countries, however, differ widely in wealth, in political stability and in how their citizens are seen abroad: high-income Gulf states, middle-income countries such as Egypt, Jordan and Morocco, and countries marked by war and displacement such as Syria, Yemen, Sudan, Libya and Somalia. We ask whether a model sees some of these nationalities more favourably than others. To put a number on this we use a task in which the nationality is always in front of the system: the money-laundering check that a bank runs before it opens an account. The model receives the bank's written policy and a customer file. The files are identical except for the nationality and the passport, and the model returns a risk rating and a recommendation.

- RQ1** Do risk ratings and recommendations differ across the 22 Arab League nationalities when the customer file is otherwise identical, and is this spread larger than for unrelated placebo nationalities? This is the main question.
- RQ2** Do the differences follow the European Union's list of high-risk countries, on which four of the 22 appear, or a country's economic and political position?
- RQ3** Are Arab League nationalities as a group rated differently from German, Turkish and placebo nationalities?
- RQ4** Do the differences remain when the policy states that nationality is not a risk factor?
- RQ5** Do the four models agree?

2 Motivation and related work

Models carry stereotypes about Arabs and Muslims. In GPT-3, 66 of 100 completions of the prompt "Two Muslims walked into a" were violent, significantly more than for other religious groups (Abid et al., 2021), and a weaker form of this association was still found in later GPT models (Hemmatian et al., 2023). In stories written by five models, characters with Middle Eastern or North African names appeared 78% less often in workplace stories than their share of the US population would suggest (Shieh et al., 2026). When models describe Middle Eastern people they disproportionately mention religion (Cheng et al., 2023). These studies treat Arabs and Muslims as one group, and they tested models released between 2020 and 2023. Our customer files state a nationality and no religion, so we measure how nationalities are treated, not bias against Islam itself.

Models also tell Arab countries apart. In text generated by GPT-2, Libyan, Sudanese and Tunisian were among the five demonyms with the most negative sentiment (Narayanan Venkit et al., 2023). In simulated immigration decisions, GPT models suggested additional scrutiny for applicants from Iraq, Sudan and Somalia, although most country effects in their choices were not significant (Mao and Zhao, 2025). Models also represent the values of some Arab countries much better than those of others (Zahraei et al., 2025). None of these studies compares the 22 nationalities with each other in a decision about a person.

Banks are a place where this matters. In a field experiment, banks in seven European countries answered 55.2% of e-mail enquiries signed with a domestic-sounding name and 31.6% of those signed with an Arabic-sounding name; for enquiries about a loan the figures were 59.8% and 35.1% (Stefan et al., 2018). Money-laundering controls have had the same effect: according to a

press report, the Netherlands Institute for Human Rights found in 2024 that a bank had discriminated against two customers whose payments it blocked and checked because of names that did not sound Dutch (NL Times, 2024). Again the evidence is about one pooled group.

In this task the nationality cannot be left out. In a CV, a nationality line can be removed before a model sees it. In a bank it cannot: German law obliges the bank to record the name, place and date of birth, nationality and address of every customer (Bundesrepublik Deutschland, 2017), and to rate the customer’s money-laundering risk. This is why we chose the task.

The rules give us a yardstick. The European guidelines tie country risk to where a customer is based or resident, where the main place of business is, and to which countries the customer has relevant links. In the 225 pages of the final report the word “nationality” does not occur, and the guidelines state that an isolated risk factor does not necessarily move a customer into a higher category (European Banking Authority, 2021). The European Commission keeps a list of high-risk third countries. Of its 26 entries, four are Arab League members: Algeria, Lebanon, Syria and Yemen (European Commission, 2026). German law applies it to relationships and transactions that involve such a country or a person resident there (Bundesrepublik Deutschland, 2017). A consumer who is legally resident in the EU must not be disadvantaged on account of nationality when opening a payment account (Bundesrepublik Deutschland, 2016), and banks should not refuse whole categories of customers without looking at the individual (European Banking Authority, 2023). For a customer who lives and works in Germany and sends no money abroad, the rules therefore give no reason to rate the passport. We can ask not only whether a model treats the 22 nationalities differently, but also whether it draws the line where the list draws it, lumps all of them together, or draws lines of its own.

AI systems are entering this task. The EU anti-money-laundering regulation, which applies from July 2027, allows decisions “resulting from automated processes [...] or from processes involving AI systems” if a decision to accept or refuse a customer is subject to “meaningful human intervention” (European Union, 2024). Vendors sell AI agents that review customer checks and screening alerts (WorkFusion, 2026; Sardine, 2026); one of

them states that ten of the twenty largest banks use its agents. These are company statements, and what such products do with a customer’s nationality is not public. For credit scoring and fraud detection, the models that banks reported to their supervisor were mainly decision trees (ECB Banking Supervision, 2025). We do not test a product; we test what open models do in the task these products are sold for.

The answer is open in both directions. In a mortgage experiment with one line stating the applicant’s race, GPT-4 Turbo recommended approval 8.5 percentage points less often for Black applicants, and an instruction to use no bias removed that gap (Bowen et al., 2025). Explicit labels can also have the opposite effect: in a French study, models recommended discrimination less often when the groups carried real origin labels than when the groups were invented (Bunel et al., 2026). We have found no study in which an LLM rates bank customers while their nationality is varied. We have not searched for one systematically yet and treat this claim as tentative.

3 Design

Materials. Everything is synthetic; the project audits model behaviour and is not a tool for checking customers. We prepared 20 customer files for a current account, with occupations from warehouse associate to software developer. Every customer lives in Germany, is employed by a German employer, expects no payments to or from other countries and has clean screening results. Ten files contain nothing unusual. Ten contain one mild indicator that the policy names, such as cash deposits from tips or an account opened by video call, so that the rating has room to move. A file carries a reference number instead of a name.

Manipulation. The nationality and the matching passport, for example Nationality: Jordanian and Jordanian passport. There are 28 conditions: the 22 Arab League nationalities, German and Turkish as benchmarks, and four unrelated nationalities as a placebo set (Uruguayan, Malawian, Maldivian, Cambodian). All non-German customers hold an unlimited residence permit. There is no version without a nationality, because the field is mandatory.

Policy. The model is told that it assists the onboarding team of a bank in Germany and that an analyst takes the final decision. It receives a one-page

Block	Runs per model
1 Policy without the paragraph on nationality	8,400
2 Policy with the paragraph on nationality	8,400
Total	16,800

Table 1: Planned runs per model: 20 files, 28 nationalities, three phrasings and five repetitions in each block.

policy that paraphrases the public rules: the risk factors, the Commission’s list of high-risk countries, a rating scale and the possible recommendations. In a second version the policy has one more paragraph, taken from the rules cited above: nationality is recorded for identification and is not a risk factor (RQ4).

Task. For each file the model returns a risk score from 0 to 100, a rating (low, medium, high), whether enhanced checks are required, and a recommendation (open, escalate to an analyst, decline). The request is phrased in three ways.

Models. Four open-weight models served with vLLM on bwUniCluster: Qwen3 8B, Llama 3.1 8B, Gemma 3 12B and Mistral Small 3.2 24B.

Size. Table 1 lists the runs per model.

4 What we will compare

RQ1 How far the risk score and the ratings differ across the 22 Arab League nationalities, against the spread across the placebo nationalities.

RQ2 The four listed countries against the eighteen others, and each nationality’s result against its country’s income group (World Bank).

RQ3 Arab League nationalities that are not on the list against the placebo nationalities and against the German and Turkish benchmarks.

RQ4 The same comparisons with the added paragraph.

RQ5 The four models against each other.

We have not fixed the statistical analysis yet. We will write down the hypotheses and the analysis before the first real run.

5 Timeline

Pitch on 13 October. A small pilot and block 1 on all four models before the midterm presentation (11–12 November). Block 2 before the final presentation (9–10 December). Report due on 8 January 2027.

Questions

1. Is a bank’s customer check a suitable setting for this course, and is the scope right for a short paper?
2. The policy we give the model names the Commission’s list of high-risk countries. Is that the right choice, or should the model receive no list?
3. Is the yardstick convincing: a customer who lives and works in Germany, for whom the rules give the passport no weight?
4. Are 20 customer files, four placebo nationalities and five repetitions enough?
5. Are four open-weight models between 8B and 24B parameters acceptable, although the products in this field run on larger commercial models?
6. The prompts are in English although the bank is German. Is that a problem?
7. Do you see concerns with reporting results by individual nationality?

AI use. The customer files, the policy text and the prompts, the literature search, and a first draft of this proposal were prepared with the help of Claude Code.

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